Introduction:

All the major components of the cells are made up of four types of macromolecules namely carbohydrates, proteins, lipids and nucleic acids. These molecules are called biomolecules. In this unit, we will see the overview of these biomolecules. Detailed information about these molecules are discussed in the subsequent chapters

2.1 Carbohydrates

Carbohydrates, otherwise known as saccharides (derived from the Greek Sakcharon– sugar; many simple sugars taste sweet) are one of the abundant molecules in earth. They are essential to maintain life in both plants and animals. They are synthesised in plants by photosynthesis.

$$x\,CO_2 + y\,H_2O \xrightarrow{sunlight} C_x(H_2O)_y + xO_2$$

The compounds that we come across in our daily life such as table sugar, wood, cotton, starch, and honey etc... are all carbohydrates.

2.1.1 Importance

Carbohydrates are widely distributed in both plant and animal tissues. They occur as food reserves in the storage organs of plants and animals. They are the important source of energy which is required for the various metabolic activities of living organisms.

They provide raw material for many important industries including textiles, artificial silks, paper, films, plastics, lacquers, confectionary, drugs, fermentation and explosives.

2.1.2 Definition

Carbohydrates are defined as poly-hydroxy aldehydes or ketones. They contain hydrogen and oxygen in the same ratio as in water (2:1)

The names of most of carbohydrates are characterised by the ending '-ose'. For example glucose, fructose, sucrose, cellulose, etc.

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